

EMAIL SECURITY

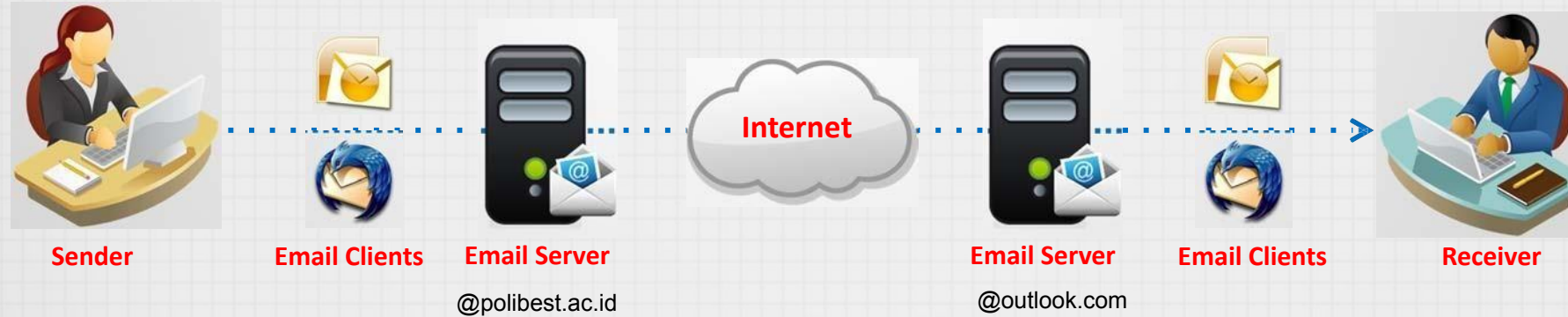
Mata Kuliah Dasar Keamanan Siber



Tri Febrianto, CEI.,CEH.,CHFI.,EHE

HOW VARIOUS EMAIL SYSTEMS WORK?

- **Email** (electronic mail) is a method of **exchanging digital messages** from a sender to one or more recipients
- Companies such as Google, Microsoft, Yahoo, etc offer free email accounts
- Email accounts can be accessed from any **web browser** or a standalone email client such as Microsoft Outlook, Mozilla Thunderbird, etc.



EMAIL SECURITY

- No email communication is **100% secure**
- Insecure emails allow attackers to **intercept personal** and **sensitive information** of the user
- If not secured, emails **sent/received** can be intercept or read by other
- Emails are one of the sources of **viruses** and various **malicious** programs
- It is necessary to **secure** emails to have **safer** communications and to protect privacy

EMAIL SECURITY THREATS

1. Spamming

The user may receive multiple email to large numbers of recipients for the purpose of commercial advertising, for the purpose of non-commercial proselytizing, for any prohibited purpose, or simply sending the same message over and over to the same user

2. Hoax/Chain Mail

The user may receive **hoax emails** that contain false information telling him/her to forward the email.

3. Phishing

Phishing mails lure victims to provide personal data

4. Malicious Email Attachments

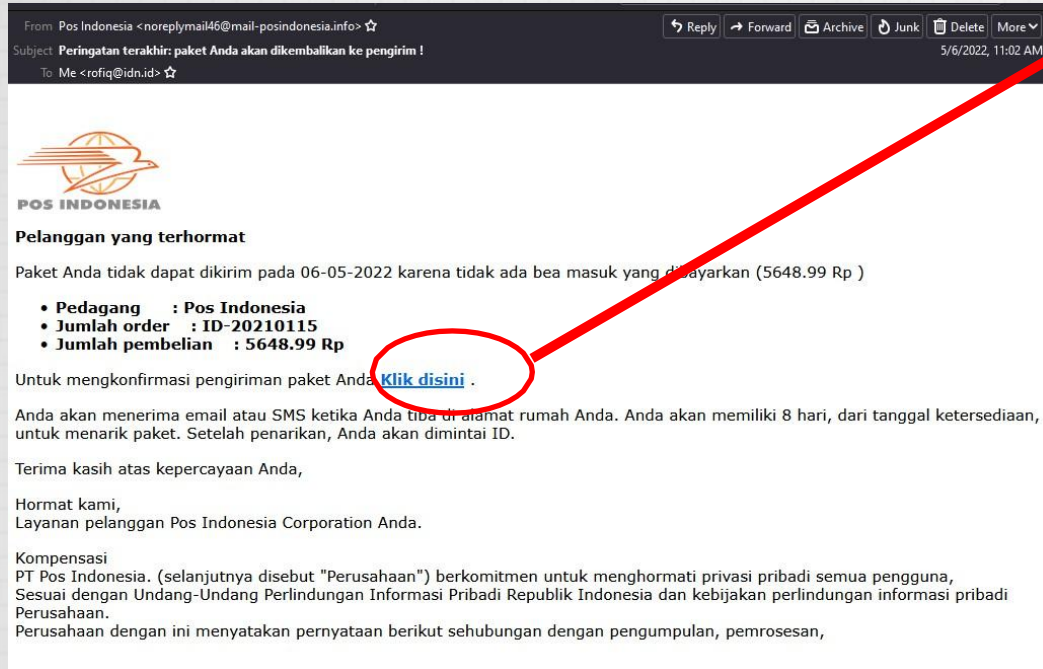
Attachments may contain a **virus**, Trojan, worms, keylogger, etc., and opening such attachments infects the computer

5. Malicious User Redirection

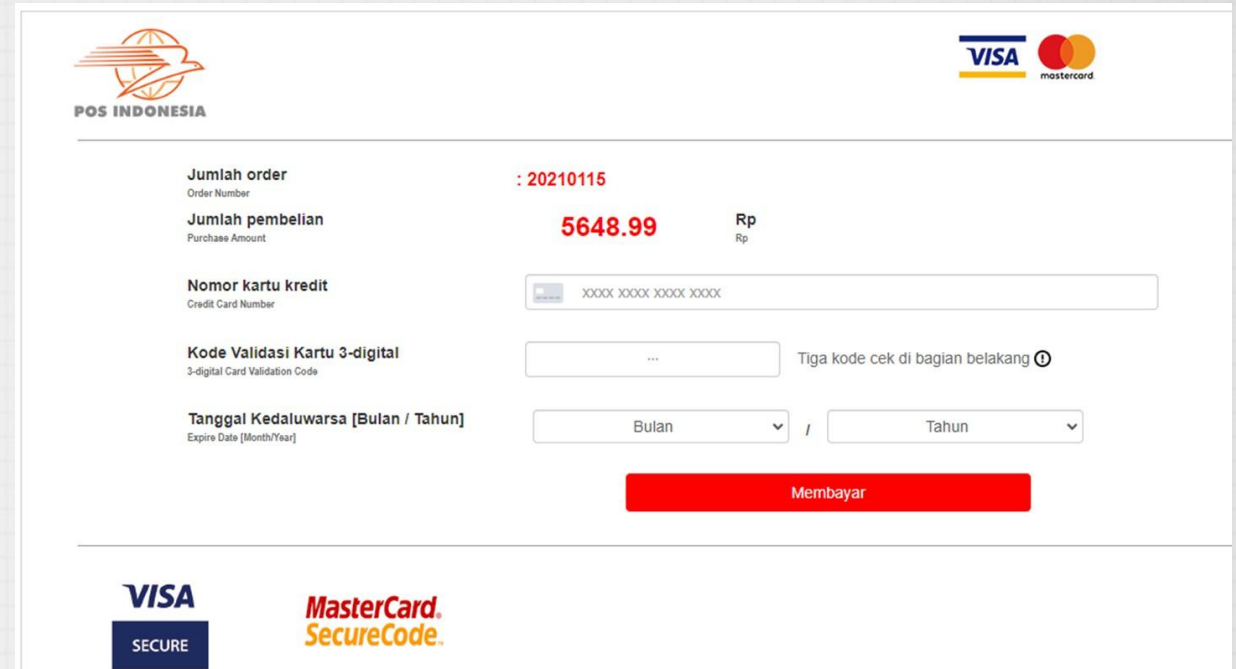
Mails may contain links that websites **hosting malwares** and pornographic material

MAIL PHISING

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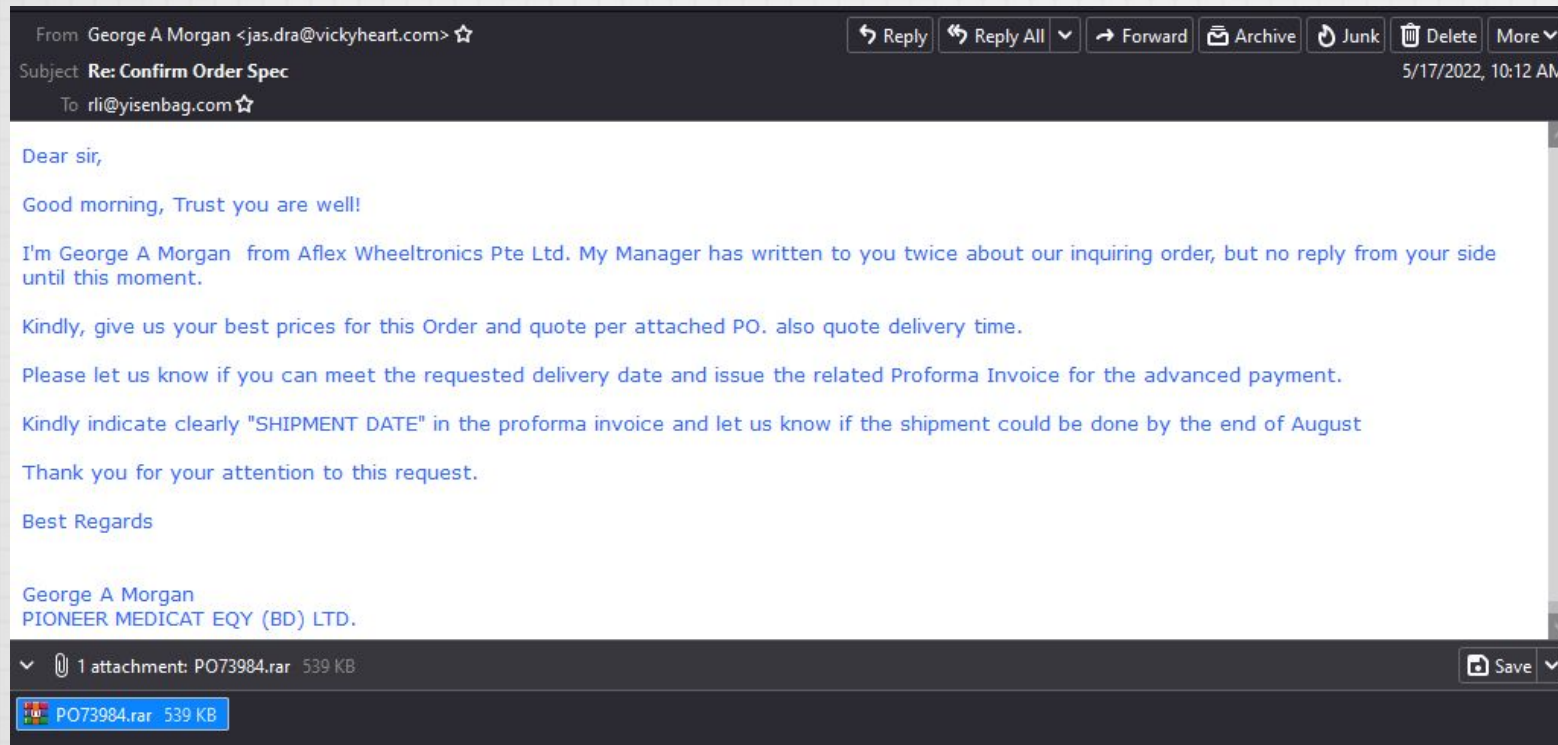


<https://indonesiapak-pos.co/pak/login/>



MALICIOUS EMAIL ATTACHMENTS

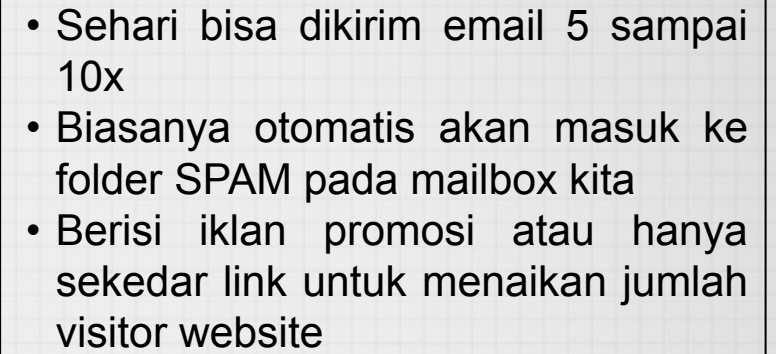
- Email attachments are major email security threats as they offers attackers easiest ways to attack a PC
- Most malicious attachments install a virus, Trojan, spyware or any other kind of malware



File tersebut apabila didownload akan dikenali sebagai malware oleh antivirus

MALICIOUS EMAIL ATTACHMENTS

- Check if the email is from one of your **contacts**
- Check if the email was ever received from the **source**
- Do not open attachments with **suspicious** or **unknown file** extensions Example: *.exe, *.vbs, *.bat, *.ini, *.bin, *.com, *.pif, *.zzx
- **Save** and **scan** all email attachments before opening them
- Check if the **subject line** and name of the **attachment** are correlated with each other
- Never open an **email attachment** from unreliable sources



SPAMMING COUNTERMEASURES

- Avoid opening spam messages (classified by spam filters)
- Report suspicious email as spam
- Use the email client's spam filter and anti- spamming tools
- Do not use official email address while registering with any website
- Never follow the links in spam messages
- Use a different email address when posting messages to any public forum

EMAIL HOAX

- Informasi yang tidak benar
- Bisa bertujuan mempengaruhi, membujuk, kampanye, atau penipuan



KANTOR PUSAT PT.WASKITA KARYA (Persero) Tbk
 JIM.Haryono Cawang Jakarta 13340 Indonesia
 Email : waskita@waskita.co.id



2. Selanjutnya peserta akan menerima bukti Reservasi, jadwal penerbangan, rincian pembayaran dan no rekening tujuan pembayaran yang di kirim pihak travel melalui alamat Email peserta (berlaku untuk peserta di luar daerah [ARODITABEK])

3. Peserta melakukan pembayaran Reservasi akomodasi/transportasi ke rekening tersebut sesuai dengan total biaya dan batas waktu yang telah ditentukan

4. Setelah peserta melakukan pembayaran, peserta mengirimkan kembali bukti pembayaran via email ke pihak Travel supaya kode E-ticket dikirim ke Email peserta untuk di print out atau dicetak sendiri. E-ticket akan di gunakan peserta sebagai tiket penerbangan setelah di bandara.

Lampiran Prosedur Penggantian Dana Transportasi:

1. Penggantian biaya transportasi Calon Karyawan berdasarkan bukti pembayaran/bukti transfer Calon Karyawan kepada Pihak PT. Duta Wahana Tour & Travel
2. Nilai nominal yang digeri oleh pihak PT Waskita Karya (Persero) Tbk adalah sesuai dengan nilai nominal yang terdapat dalam bukti transfer
3. Calon Karyawan menyerahkan bukti pembayaran/bukti transfer kepada Pihak PT. Duta Wahana Tour & Travel setelah di tempat pelaksanaan kegiatan
4. Pihak PT. Duta Wahana Tour & Travel mengajukan permintaan pembayaran atas tagihan biaya transportasi kepada Direktur Keuangan PT Waskita Karya (Persero) Tbk
5. Direktur Keuangan PT Waskita Karya (Persero) Tbk membayar tagihan kepada Pihak PT. Duta Wahana Tour & Travel
6. Pihak PT. Duta Wahana Tour & Travel melaksanakan penggantian pembayaran sesuai nilai nominal transfer kepada Calon Karyawan sebelum acara dimulai

Tugas Dan Tanggung Jawab Apes Perjalanan (PT. Duta Wahana Tour & Travel)

1. Mengajukan Pengajuan (Hotel) dan Memesan tiket transportasi Calon Karyawan (keberangkatan dan kedatangan Calon Karyawan)
2. Mengantar Calon Karyawan(s) dari Bandar Udara (airport) Jakarta ke tempat pelaksanaan kegiatan
3. Mengantar Calon Karyawan(s) untuk mobilisasi (pengalangan) dan sampai kegiatan ke Bandar Udara (airport) Jakarta

Peserta Yang Lolos

1. Setiap peserta yang lulus seleksi akan kami hubungi (Via Phone/Telp).
2. Seleksi masuk tidak dipungut biaya.
3. Peserta yang dinyatakan lulus setelah wajib menyerahkan pas foto terbaru berwarna dengan ukuran 4x6 sebanyak 2 lembar.
4. Apabila Pelamar memberikan keterangan data tidak benar baik pada setiap tahapan seleksi, maupun setelah menjadi karyawan, maka kami berhak membatalkan keputusan tersebut dan atau membereskan sebagai karyawan di perusahaan.

Kebijakan PT.Waskita Karya Persero Tbk

PT. Waskita Karya Persero Tbk menetapkan gaji karyawan yang di dasarkan pada level tuntutan serikat pekerja serta nantinya akan di tempatkan di kantor cabang PT. Waskita Karya Persero Tbk.

Demikian penjelasan singkat yang dapat kami sampaikan dan kami selaku panitia seleksi recruitment PT. Waskita karya persero Tbk menyampaikan terima kasih kepada semua pihak atas atensi dan partisipasinya dan Bismillah Rekrutmen akan dilaksanakan apabila dalam proses seleksi ada hal yang tidak

CAPTURING EMAIL COMMUNICATION

1
1

⏮ Reply all | ⏶ | 🗑 Delete | Not junk ⏷ | Block sender | ⋮

Data Penting

- ⓘ This message was identified as junk. We'll delete it after 30 days. It's not junk
- ⓘ Retention: Junk Email (30 days) Expires: Fri 6/17/2022 7:21 AM

 **admin@indomedia.net.id**
To: Rofiq Fauzi

      
Wed 5/18/2022 7:21 AM

Ini data username dosen

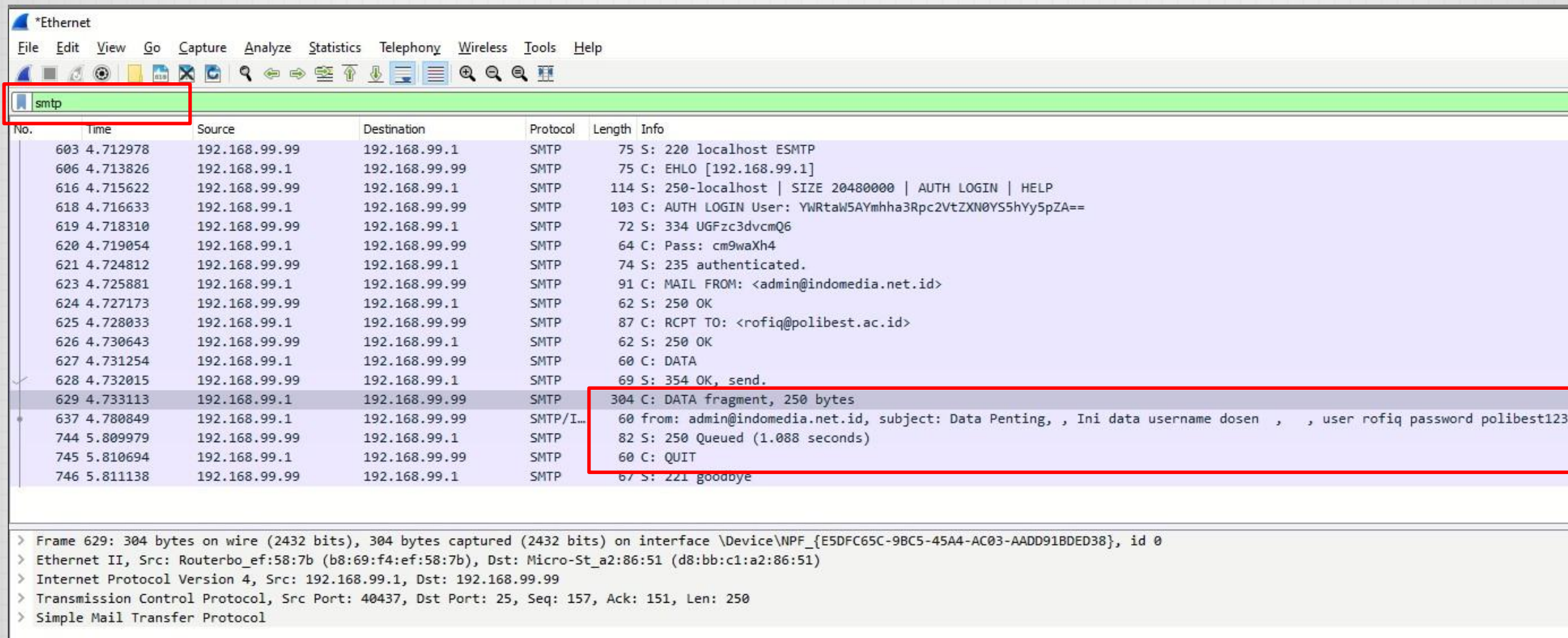
user rofiq password polibest123

⏮ Reply

⏭ Forward

CAPTURING EMAIL COMMUNICATION

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*Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

smtp

No.	Time	Source	Destination	Protocol	Length	Info
603	4.712978	192.168.99.99	192.168.99.1	SMTP	75	S: 220 localhost ESMTP
606	4.713826	192.168.99.1	192.168.99.99	SMTP	75	C: EHLO [192.168.99.1]
616	4.715622	192.168.99.99	192.168.99.1	SMTP	114	S: 250-localhost SIZE 20480000 AUTH LOGIN HELP
618	4.716633	192.168.99.1	192.168.99.99	SMTP	103	C: AUTH LOGIN User: YWRtaW5AYmhma3Rpc2VtZXN0YS5hYy5pZA==
619	4.718310	192.168.99.99	192.168.99.1	SMTP	72	S: 334 UGFzc3dvcmQ6
620	4.719054	192.168.99.1	192.168.99.99	SMTP	64	C: Pass: cm9waXh4
621	4.724812	192.168.99.99	192.168.99.1	SMTP	74	S: 235 authenticated.
623	4.725881	192.168.99.1	192.168.99.99	SMTP	91	C: MAIL FROM: <admin@indomedia.net.id>
624	4.727173	192.168.99.99	192.168.99.1	SMTP	62	S: 250 OK
625	4.728033	192.168.99.1	192.168.99.99	SMTP	87	C: RCPT TO: <rofiq@polibest.ac.id>
626	4.730643	192.168.99.99	192.168.99.1	SMTP	62	S: 250 OK
627	4.731254	192.168.99.1	192.168.99.99	SMTP	60	C: DATA
628	4.732015	192.168.99.99	192.168.99.1	SMTP	69	S: 354 OK, send.
629	4.733113	192.168.99.1	192.168.99.99	SMTP	304	C: DATA fragment, 250 bytes
637	4.780849	192.168.99.1	192.168.99.99	SMTP/I...	60	from: admin@indomedia.net.id, subject: Data Penting, , Ini data username dosen , , user rofiq password polibest123
744	5.809979	192.168.99.99	192.168.99.1	SMTP	82	S: 250 Queued (1.088 seconds)
745	5.810694	192.168.99.1	192.168.99.99	SMTP	60	C: QUIT
746	5.811138	192.168.99.99	192.168.99.1	SMTP	67	S: 221 goodbye

> Frame 629: 304 bytes on wire (2432 bits), 304 bytes captured (2432 bits) on interface \Device\NPF_{E5DFC65C-9BC5-45A4-AC03-AADD91BDED38}, id 0

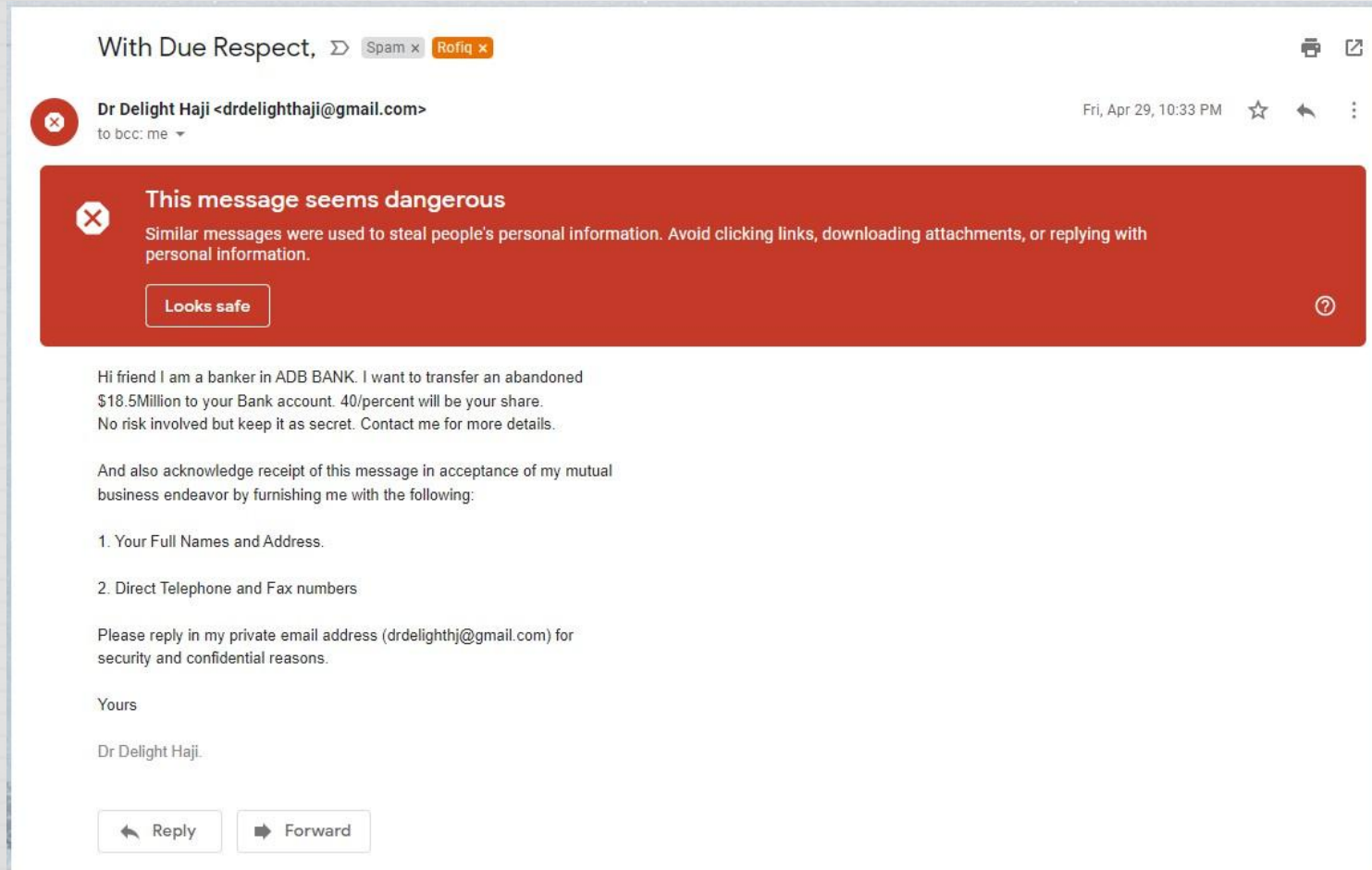
> Ethernet II, Src: Routerbo_ef:58:7b (b8:69:f4:ef:58:7b), Dst: Micro-St_a2:86:51 (d8:bb:c1:a2:86:51)

> Internet Protocol Version 4, Src: 192.168.99.1, Dst: 192.168.99.99

> Transmission Control Protocol, Src Port: 40437, Dst Port: 25, Seq: 157, Ack: 151, Len: 250

> Simple Mail Transfer Protocol

NIGERIAN SCAM



A Nigerian scam is a form of **advance payment of money** or money transfer

This scam is called a Nigerian scam because initially it started from Nigeria, but they can come in anywhere in the world

Using this scam, scammers contact you by **sending an email** and offer you a share in a large sum of money

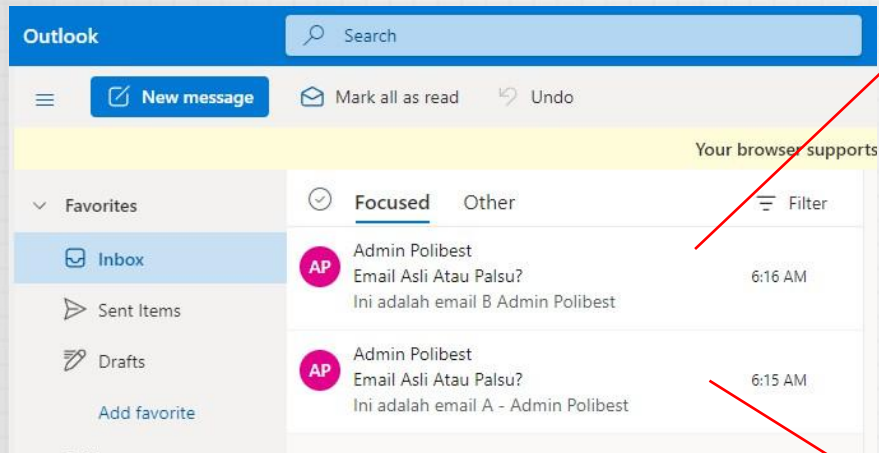
They say they want to **transfer money**, which was trapped in banks during civil wars, to your account

They may also cite various reasons such as massive inheritance problems, government restrictions, or taxes in the scammer's country

Scammers ask you to pay money or give them your **bank account details** to help them transfer the money

EMAIL IDENTIFICATION

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Email Asli Atau Palsu?



Admin Polibest <admin@bhaktisemesta.ac.id>

To: You



Fri 5/20/2022 6:15 AM

Ini adalah email A

Admin Polibest

Reply

Forward

Email Asli Atau Palsu?



Admin Polibest <admin@bhaktisemesta.ac.id>

To: You



Fri 5/20/2022 6:16 AM

Ini adalah email B

Admin Polibest

Reply

Forward

EMAIL IDENTIFICATION

Message source

Received: from [127.0.0.1] (DOSENRS [127.0.0.1])
by localhost with ESMTTPA
; Fri, 20 May 2022 06:16:00 -0700

Message-ID: <a81e38e4-5736-ceb1-2f44-34aa40562120@bhaktisemesta.ac.id>
Date: Fri, 20 May 2022 06:16:00 -0700
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:91.0) Gecko/20100101
Thunderbird/91.9.0
Reply-To: rofiq.fauzi@gmail.com
Content-Language: en-US
To: dosenrks@outlook.com
From: Admin Polibest <admin@bhaktisemesta.ac.id>
Subject: Email Asli Atau Palsu?
Content-Type: text/plain; charset=UTF-8; format=flowed
Content-Transfer-Encoding: 7bit
X-IncomingHeaderCount: 12
Return-Path: admin@bhaktisemesta.ac.id
X-MS-Exchange-Organization-ExpirationStartTime: 19 May 2022 23:16:14.9492
(UTC)
X-MS-Exchange-Organization-ExpirationStartTimeReason: OriginalSubmit
X-MS-Exchange-Organization-ExpirationInterval: 1:00:00:00.0000000
X-MS-Exchange-Organization-ExpirationIntervalReason: OriginalSubmit

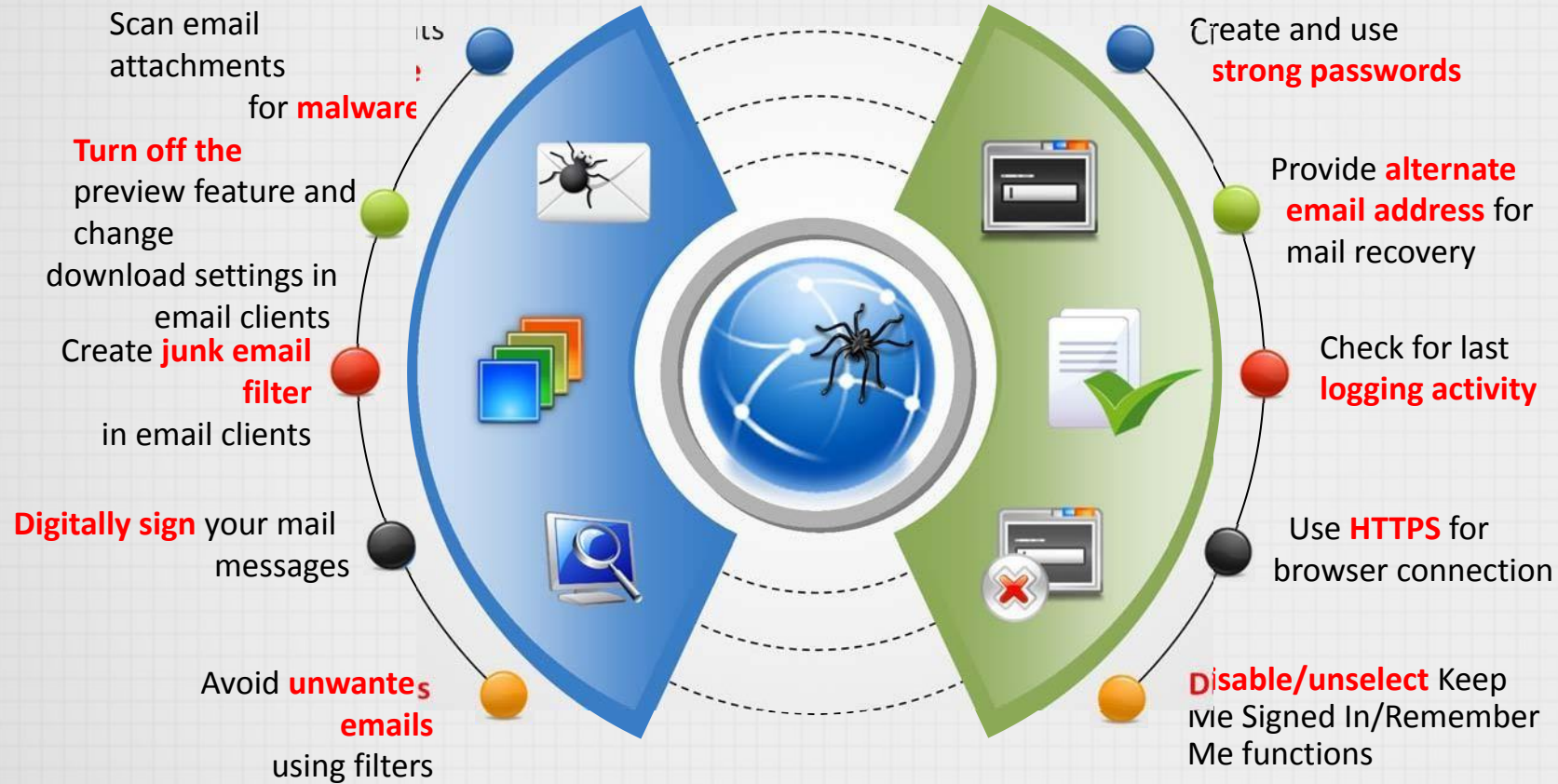
Close

Message source

Received: from localhost (Unknown [127.0.0.1])
by kirim.cloudhost.id (Haraka/2.8.27) with ESMTTPSA id 849EBD08-FD71-4584-9F13-CCD6B7650726.1
envelope-from <admin@bhaktisemesta.ac.id>
tls TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (authenticated bits=0);
Fri, 20 May 2022 06:15:02 +0700
Received: from [127.0.0.1] (port=45652 helo=iix11.cloudhost.id)
by iix11.cloudhost.id with esmtpp (Exim 4.95)
(envelope-from <admin@bhaktisemesta.ac.id>)
id 1nrpMC-0002Xi-TW
for dosenrks@outlook.com;
Fri, 20 May 2022 06:15:02 +0700
X-Originating-IP: 103.129.220.144
X-Spampanel-Domain: cloudhost.id
X-Spampanel-Username: iix18
Authentication-Results-Original: antispamcloud.com; auth=pass (plain)
smtp.auth=iix18@cloudhost.id
X-Spampanel-Outgoing-Class: unsure
X-Spampanel-Outgoing-Evidence: Combined (0.82)
X-Recommended-Action: accept
X-Filter-ID: Pt3MvcO5N4iKaDQ5O6lkdGIMVN6RH8bjRMzItlySaT9uccVTBIYt+VagxqTJJN/iPUtbdvnXkkgZ
3YnVld/Y5jcf0yeVQAvfjHznO7+bT5whSTiIUPYZSo21PDqnetmal4T/zhoa6uhUwCFnbVxRNh4

Close

EMAIL SECURITY PROCEDURES



KEEP ME SIGNED IN/REMEMBER

- Most of the popular email clients have the **Keep me signed in** or **Remember Me** options
- Checking these options allow the email client to fetch the email inbox of the user without him/her having to fill in the login details again
- This allows other users to access the user's email
- Users should check that this option is **not selected** when accessing email from a public computer

The image displays three different login interfaces, each with a checkbox option highlighted by a red rectangle to illustrate the concept of 'Keep me signed in' or 'Remember Me'.

- Top Left (Yahoo!):** The form is titled 'Sign in to Yahoo!'. It includes fields for 'Yahoo! ID' and 'Password'. Below the password field is a checkbox labeled 'Keep me signed in (Uncheck if on a shared computer)'. A 'Sign In' button is located below the checkbox.
- Top Right (Google):** The form is titled 'Sign in with your Google Account'. It includes fields for 'Username' and 'Password'. Below the password field is a checkbox labeled 'Stay signed in'. A 'Sign in' button is located below the checkbox.
- Bottom Right:** The form is titled 'Sign In'. It includes fields for 'E-mail or Screen Name' and 'Password'. Below the password field is a checkbox labeled 'Remember Me'. A 'Sign In' button is located to the right of the checkbox.